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Department of Computer Science and Engineering
Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: III Semester B.E. CSE 'A'
Course Code & Title: CS3391 Object Oriented Programming
Name of the Faculty member (s): Mrs.B.Vijayalakshmi, AP (SG)/CSE

Innovative Practice Description

- **Unit / Topic: Unit IV / String & String Buffer**
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO10
- **Activity Chosen: Collaborative learning**
- **Justification:**
  The concept of string and string buffer are important concepts while designing any application. The understanding of proper usage of string, string buffer and its methods are essential. Hence a collaborative learning methodology is chosen to make the students to understand it clearly.
- **Time Allotted for the Activity: 45 Minutes**
- **Details of the Implementation:**
  - The collaborative learning activity provide students the best opportunities to practice the tasks in an authentic way, including interactions and working with other people.
  - The activity was conducted for the duration of 45 minutes.
  - Initially the teams are formed based on self selection in which the students are form their own groups of size 5. Totally 11 teams are formed.
  - Once the teams are formed, they are assigned with a particular task on strings, string buffer and its methods.
  - The team members started working on their assigned task and discussed with them. It is shown in Fig.1.
  - The instructor monitored all the teams and provided any clarification if needed.
  - After 20 minutes of discussion the faculty member called every team to present their findings regarding the posted questions.
  - The team members explained the methods with examples. Fig.2 shows the presentation by Mr. Ramkumar from team 1.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO4	3	3	2	1	2	2	1	3	1	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO9	PO10	PSO1	PSO2	PSO3
	3	**2**	**1**	**1**	**3**	**1**	**1**
Justification for correlation	The students apply the knowledge of strings	The students analyze and apply appropriate string methods depending upon the situation.	The students work as a team to find solutions.	The students communicate the problem solutions effectively to their team members	The strings and methods are used in developing the software	The concept is used in security applications	The concept is used in providing AI based solutions

- **Images / Screenshot of the practice:**

Fig. 1: Team Discussion

Fig. 2: Presentation by Mr.Ramkumar from Team 1

- **Reflective Critique:**
  - ❖ ***Feedback of practice from students and other stakeholders:***
    - All the teams actively participated in the activity. The students interact with each other and eagerly learn from their team.
  - ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
    - This activity helps the students to get more clarity on the string methods.
    - It helps the students are made to work in a team and share their ideas to
    - others and also improve their communication skills
    - The misunderstood concepts are clarified while the team members are presenting.
  - ❖ ***Challenges faced in implementation:***
    - It takes more time to make all the teams present the answers to the given questions.
    - Some of the team members hesitate to present in front of the class.

References:

- https://www.monash.edu/learning-teaching/TeachHQ/Teaching-practices/collaborative-learning/how-to/collaborative-learning-processes
- https://www.ritrjpm.ac.in/images/computer-science/2022-2023/Unit_2_collaborative%20learn.pdf